

BML lecture #7: Simulation-based Inference

<http://github.com/rbardenet/bml-course>

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- ▶ Part I (Last week): Using generative models to solve (Bayesian) inverse problems.
- ▶ Part II (Today): Simulation-based / Likelihood Free inference.

1 Approximate Bayesian computation

2 Neural-based SBI

- ▶ Prior $\pi \in \mathcal{P}_1(\Theta)$.
- ▶ Simulator $L : \Theta \ni \theta \rightarrow L_\theta \in \mathcal{P}_1(\mathcal{Y})$.

Algorithm 1: Posterior sampling from Rubin, 1984.

Data: $n \geq 0, y_\star \in \mathcal{Y}$.

Result: $\theta_{1:\tau}$

$\hat{\theta}_{1:n} \sim \pi^{\otimes n};$

$Y_{1:n} \sim L_{\hat{\theta}_1} \otimes \cdots \otimes L_{\hat{\theta}_n};$

$\tau \leftarrow 0;$

for $i \in \{1, \dots, n\}$ **do**

if Y_i matches Y **then**

$\tau = \tau + 1;$

$\theta_\tau \leftarrow \hat{\theta}_i;$

end

end



Figure: Donald B. Rubin

What does matching means?

In general

$$y \text{ matches } y_{\star} \text{ means } d(S(y), S(y_{\star})) < \epsilon. \quad (1)$$

Algorithm 2: Vanilla Approximate Bayesian Computation (ABC) algorithm.

Data: $n \geq 0$, $y_{\star} \in \mathcal{Y}$, $\epsilon > 0$.

Result: $\theta_{1:\tau}^{\epsilon}$

$\hat{\theta}_{1:n} \sim \pi^{\otimes n};$

$Y_{1:n} \sim L_{\hat{\theta}_1} \otimes \cdots \otimes L_{\hat{\theta}_n};$

$\tau \leftarrow 0;$

for $i \in \{1, \dots, n\}$ **do**

if $d(S(Y_i), S(y_{\star})) < \epsilon$ **then**

$\tau = \tau + 1;$

$\theta_{\tau}^{\epsilon} \leftarrow \hat{\theta}_i;$

end

end

`https://codeberg.org/gvc/
tutorial_sbi`

$$p_{\epsilon}(A|y_{\star}) := \frac{\int 1_A(\theta) 1_{B_{\epsilon}(S(y_{\star}))}(y) \ell(dy|\theta) \pi(d\theta)}{\int 1_{B_{\epsilon}(S(y_{\star}))}(\tilde{y}) \ell(d\tilde{y}|\tilde{\theta}) \pi(d\tilde{\theta})}. \quad (2)$$

$$p(d\theta|y) := \frac{\ell(y|\theta) \pi(d\theta)}{\int \ell(y|\tilde{\theta}) \pi(d\tilde{\theta})} \quad (3)$$

What can we say about the convergence of $p_{\epsilon}(\cdot|y_{\star})$ to $p(\cdot|y)$? Under which conditions on $S()$?

Two sources of randomness to be dealt with: L and π .

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We focus on the work developed in

- ▶ D. T. Frazier et al. “Asymptotic Properties of Approximate Bayesian Computation”. In: *Biometrika* 105.3 (2018), pp. 593–607. JSTOR: 48546332. URL: <https://www.jstor.org/stable/48546332> (visited on 02/23/2026),
- ▶ David T. Frazier, Christian P. Robert, and Judith Rousseau. “Model Misspecification in Approximate Bayesian Computation: Consequences and Diagnostics”. In: *Journal of the Royal Statistical Society Series B (Statistical Methodology)* 82.2 (Apr. 1, 2020), pp. 421–444. URL: <https://doi.org/10.1111/rssb.12356> (visited on 02/23/2026).

In it, $y = (y_1, \dots, y_T)$ and $S(y)$ should concentrate as T grows !

Condition A_0 .

There exists a (non-random) $b : \Theta \rightarrow \mathcal{S}$ and a sequence of monotone non-increasing functions $\rho_T : \mathbb{R}_* \rightarrow \mathbb{R}_*$ such that $\lim_{n \rightarrow \infty} \rho_n(u) = 0$ and that for all $\theta \in \Theta$

$$1 - \mathbb{P}_{L_\theta} [1_{B_u(b(\theta))}(S(Y))] \leq c(\theta)\rho_T(u), \quad \int c(\theta)\pi(d\theta) < \infty. \quad (4)$$

Furthermore, there exists a positive sequence $v_T \rightarrow \infty$ and $u_0 \in \mathbb{R}_*$ such that either

Poly There exists $\kappa \in \mathbb{R}_*$ such that for $u \leq u_0$, $\rho_T(u) = (v_T u)^{-\kappa}$,

Exp There exists $h_\theta : \mathbb{R}_* \rightarrow \mathbb{R}_*$, $(m, C, \tau) \in \mathbb{R}_*^3$ such that c
 $\rho_T(u) = \exp(-h_\theta(uv_T))$ and for $u \leq u_0$

$$\int c(\theta) \exp(-h_\theta(uv_T)) \pi(d\theta) \leq C \exp(-m(uv_T)^\tau). \quad (5)$$

In words: $S(Y)$ concentrates over $b(\theta)$ under L_θ .

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If we have a statistics of the type

$$S(y) = T^{-1} \sum_{i=1}^T g(y_i), \quad (6)$$

with $y_1, \dots, y_T$ iid and take $b(\theta) = \mathbb{E}_{L_\theta} [S(y_1)]$, provided that $\mathbb{E} [|g(y_1) - b(\theta)|^p] < \infty$, we have by Markov Inequality that

$$\mathbb{P}_{L_\theta} [|b(\theta) - S(Y)| \geq u] \leq \mathbb{E}_{L_\theta} [|b(\theta) - g(y_1)|^p] / (u^p T^{p-1}). \quad (7)$$

Condition A_1 .

There exists a unique b_0 such that for all $C > 0$,
 $\lim_{T \rightarrow \infty} \mathbb{P}_0 [d(S(Y), b_0)] = 0$ and there exists $v_{0,T} \rightarrow \infty$ such that

$$\liminf_T 1 - \mathbb{P}_0 \left[1_{B_{1/v_{0,T}}(b_0)}(S(Y)) \right] = 1. \quad (8)$$

We define

$$\Theta_\star := \arg \min_{\theta \in \Theta} d(b(\theta), b_0) .$$

We also define

$$\epsilon_\star = \inf_{\theta \in \Theta} d(b(\theta), b_0) .$$

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Condition A_2

There exists $(D, M_0, \delta_0) \in \mathbb{R}_*^3$ such that for all $\delta_0 \geq \delta > 0$ and $M \geq M_0$, there exists $S_\delta \subset \{\theta \in \Theta \mid d(b(\theta), b_0) - \epsilon_* \leq \delta\}$ such that

Poly $\int_{S_\delta} \left(1 - \frac{c(\theta)}{M}\right) \pi(d\theta) \geq \tilde{C} \delta^D,$

Exp $\int_{S_\delta} (1 - c(\theta) \exp(-h_\theta(M))) \pi(d\theta) \geq \tilde{C} \delta^D.$

Corollary 1 David T. Frazier, Robert, and Rousseau, 2020 (Informal version)

Assume A_0, A_1, A_2 . If $\epsilon_T \rightarrow \epsilon$ and $\epsilon_t \geq \epsilon_\star + Mv_T^{-1} + v_{0,T}^{-1}$ then for any $\delta > 0$ and any $C > 0$

$$\lim_T \mathbb{P}_0 [p_{\epsilon_T} (\{\theta \mid d(\theta, \Theta_\star) \geq \delta\} | y) \geq C] = 0 \quad (9)$$

- ▶ MCMC ABC: Marjoram et al., 2003
- ▶ SMC ABC: Sisson, Fan, and Tanaka, 2007
- ▶ Post ABC recalibration: Rodrigues, Prangle, and Sisson, 2018.
- ▶ Neural Summaries: Huang et al., 2023
- ▶ ABC for IID data via Statistical distances: Gutmann et al., 2018; Bernton et al., 2019.

NON EXHAUSTIVE ! Go check online for more.

1 Approximate Bayesian computation

2 Neural-based SBI

Here we followed the first "modern" kind: Papamakarios, Pavlakou, and Murray, 2017. General approach:

- ▶ Simulate a dataset $\mathcal{D} = \{(\theta_i, Y_i) | i \in \{1, \dots, N\}\}$, according to the joint $A \rightarrow \int 1_A(\theta, y) L_\theta(dy) \pi(d\theta)$.
- ▶ Learn a "neural" generative model $q_\phi(\cdot|y)$ to approximate $p(\cdot|y)$.

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Flow layer

For a given parameter ϕ , a flow layer is defined as

$$T_{\phi} : \mathbb{R}^{d_y} \rightarrow \mathbb{R}^{d_y} ,$$

such that T_{ϕ} is a diffeomorphism with *known inverse* T_{ϕ}^{-1} and *easy to calculate* $\det(J_{T_{\phi}^{-1}})$.

Example (Real NVP):

$$T_{\phi}(x_0, x_1) = (x_0, \exp(n_{\phi}(x_0))x_1 + \mu_{\phi}(x_0)) ,$$

where n_{ϕ}, μ_{ϕ} are MLP networks.

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Normalizing flow

A normalizing flow is simply defined by the sequential procedure $Z_0 \sim \lambda$

$$Z_i = P_i^{-1} T_{\phi_i} P_i(Z_{i-1}) . \quad (10)$$

where P_i is a simple permutation of the coordinates. (For example $P_i(x_0, x_1) = (x_1, x_0)$ if i is odd else $P_i = \text{Id.}$)

The density is

$$p_\phi(x) = \lambda(T_{\phi_1}^{-1} \cdots T_{\phi_k}^{-1}(x)) \prod_{i=1}^k \det T_{\phi_i}^{-1}(T_{\phi_{i+1}}^{-1} \cdots T_{\phi_k}^{-1}(x)) \quad (11)$$

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$$T_{\phi}(x_0, x_1|y) = (x_0, \exp(n_{\phi}(x_0, y))x_1 + \mu_{\phi}(x_0, y)) ,$$

$$\phi_{\star} \in \arg \min_{\phi} \mathbb{E}_{\theta \sim \pi} [\mathbb{E}_{Y \sim L_{\theta}} [-\log q_{\phi}(\theta|Y)]] \quad (12)$$

Note that

$$\mathbb{E}_{\theta \sim \pi} [\mathbb{E}_{Y \sim L_{\theta}} [-\log q_{\phi}(\theta|Y)]] = \mathbb{E}_{Y \sim p_Y} [\mathbb{E}_{\theta \sim p(\theta|Y)} [-\log q_{\phi}(\theta|Y)]] \ .$$

Furthermore

$$\begin{aligned} \arg \min_{\phi} \mathbb{E}_{Y \sim p_Y} [\mathbb{E}_{\theta \sim p(\theta|Y)} [-\log q_{\phi}(\theta|Y)]] \\ = \arg \min_{\phi} \mathbb{E}_{Y \sim p_Y} [\text{D}_{\text{KL}}(p(\cdot|Y) || q_{\phi}(\cdot|Y))] \ . \end{aligned}$$

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- ▶ SNPE: Greenberg, Nonnenmacher, and J. Macke, 2019; Radev et al., 2022; Deistler, Goncalves, and J. H. Macke, 2022
- ▶ Score Based NPE: Geffner, Papamakarios, and Mnih, 2023; Sharrock et al., 2024
- ▶ Flow Matching NPE: Wildberger et al., 2023

Imagine you have n observations $y_1, \dots, y_n$. What do you do?

Learn a "proxy":

$$q_{\phi}(\cdot|\theta) \approx L_{\theta} \quad (13)$$

using a normalizing flow:

$$\arg \min_{\phi} \mathbb{E}_{\theta \sim \pi} [\mathbb{E}_{Y \sim L_{\theta}} [-\log q_{\phi}(Y|\theta)]] . \quad (14)$$

Any MCMC sampler targeting

$$q_{\phi}(y|\theta)\pi(\theta) .$$

What about sampling from n observations?

Target:

$$\prod_{i=1}^n q_{\phi}(y_i|\theta)\pi(\theta) .$$

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Problem: What if simulation is too costly to generate a "big enough" dataset to have a good approximation for all θ ? (or at least with high probability under π ?)

For a given y , we might want to learn only around $p(\theta|y_*)$, i.e.,

$$\arg \min_{\theta} \mathbb{E}_{\theta \sim p(\cdot|y_*)} [\mathbb{E}_{Y \sim L_{\theta}} [-\log q_{\phi}(Y|\theta)]]$$

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Algorithm 3: Sequential Learning and sampling from Papamakarios, Sterratt, and Murray, 2019.

Data: $R \geq 1$, $y_\star \in \mathcal{Y}$, $n \in \mathbb{N}_\star$, pretrained $q_{\phi^0}(\cdot)$.

Result: $\theta_{1:n}^R$

$\mathcal{D} \leftarrow \{\};$

for $r \in \{1, \dots, R\}$ **do**

$\hat{\theta}_{1:n}^r \sim q_{\phi^{r-1}}(\theta|y_\star)\pi(\theta) / Z(y_\star, \phi^{r-1})$ via MCMC ;

$Y_{1:n}^r \sim L_{\hat{\theta}_1^r} \otimes \dots \otimes L_{\hat{\theta}_n^r};$

$\mathcal{D} \leftarrow \mathcal{D} \cup \{(\hat{\theta}_i^r, Y_i^r) | i \in \{1, \dots, n\}\};$

$\phi^r \in \arg \min_{\phi} \sum_{(\theta, Y) \in \mathcal{D}} -\log q_{\phi}(\theta|Y);$

end

Bayes Classifier:

$$p(y = 1|x) = \frac{p_1(x)}{p_1(x) + p_0(x)} = \frac{r(x)}{r(x) + 1} , \quad (15)$$

where $r(x) = p_1(x)/p_0(x)$.

Hermans, Begy, and Louppe, 2020 proposes approximating

$$r(y|\theta, \tilde{\theta}) = \frac{\ell(\theta|y) \pi(\theta)}{p(y) \pi(\hat{\theta})} = \frac{p(\theta|y)}{\pi(\hat{\theta})} , \quad (16)$$

by learning a classifier to distinguish $\ell(\theta|y) \pi(\theta)$ from $p(y) \pi(\hat{\theta})$.

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- ▶ SBI Package
- ▶ SBI benchmark
- ▶ Tutorial: Deistler, Boelts, et al., 2025

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